# Eric Xu

*Curriculum Vitae* | *November 2026*

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## Research Interests

Scientific machine learning; rare-event sampling and AI ensemble forecasting of weather extremes; data assimilation; dynamical systems; GPU computing for large AI weather models.

## Education

**University of California, Santa Cruz** — Santa Cruz, CA

*M.S. in Scientific Computing and Applied Mathematics, Plan I (thesis)* — *Expected June 2027*

- One-year accelerated track. Advisor: Prof. Ashesh Chattopadhyay (TACS Lab).

*B.S. in Applied Mathematics* — *2024 – 2026*

- Highest Honors in the Major. GPA 3.77/4.00. Dean's Honors (4 quarters).

## Research Experience

**TACS Lab, UC Santa Cruz** — Santa Cruz, CA

*Researcher, B.S. and M.S. (Advisor: Prof. Ashesh Chattopadhyay)* — *Sep 2025 – Present*

- Adapt AI+RES rare-event sampling (Lancelin et al., 2025) to subseasonal forecasting: AI weather-model ensembles (FourCastNet 3, GenCast) are cloned and pruned by score to sample heat waves and cold outbreaks 3 to 4 weeks ahead.
- Built the batch pipeline (JAX, PyTorch; Slurm on H100 nodes, PBS on NCAR Derecho) that launches ensembles weeks before past US extremes and checks against ERA5 (0.25°) whether they reach the observed event.

**Vayuh.ai**

*ML Researcher (part-time)* — *Jul 2026 – Present*

- Cut GenCast's 0.25° JAX executable from 4.17 GB to 107 MB so it fits the persistent compile cache, and added persistent GPU workers, so each forecast case compiles the model once instead of at every leg.
- Benchmarked GenCast, FourCastNet 3, and SFNO as ensemble members on H100 nodes, then moved the ensemble to FourCastNet 3 walkers with an SFNO scorer: about 27x less compute per forecast case (33.5 to 1.2 H100-hours).

## Selected Projects

**4D-Var Data Assimilation with a Learned Inverse Operator** | *PyTorch, L-BFGS* — Apr – May 2026

- AM 170B capstone (team of 5): wrote most of the PyTorch port of Frerix et al. (ICML 2021, originally JAX), reproducing the learned inverse observation operator and its Lorenz-96 results.
- Extended the paper with the full 4D-Var cost, noisy observations, and sliding-window cycling; studied gradient blow-up over long assimilation windows and ran the sliding-window experiments on Lorenz-96 and 2D Kolmogorov flow.
- Code: github.com/ericxu1070/invobs-data-assimilation

**Cross-Species Transfer Learning with DeepSEA** | *PyTorch, scikit-learn* — Jun 2026

- Fine-tuned the human-trained DeepSEA CNN on mouse ENCODE DNA windows. It beat training from scratch on 115 of 133 chromatin features (median AUROC 0.877 vs 0.865) and converged in fewer epochs.
- Showed that zero-shot transfer to mouse scores near chance (median AUROC 0.50), so the model needs fine-tuning.
- Code: github.com/ericxu1070/DeepSEA-Transfer-Learning

**Council of Sumer** | *Python, FastAPI, AGNTCY SLIM, OpenTelemetry* — Oct 2026

- Built a council of five LLM agents with conflicting goals (hackathon, team of 3): they run as separate processes, debate over AGNTCY SLIM, and vote under runoff and human tie-break rules to govern the city in Hamurabi.
- Co-wrote the deliberation engine and benchmark: over 16 seeded games the council improved on the single-LLM baseline.
- Code: github.com/ericxu1070/council-of-sumer

## Teaching

**Baskin School of Engineering, UC Santa Cruz** — Santa Cruz, CA

*Teaching Assistant* — *Fall 2026*

- AM 20 Ordinary Differential Equations for Engineers; AM 11A Mathematical Methods for Economists I.
- Run multiple weekly group tutoring sessions and office hours, and grade assignments.
- Pedagogy training: AM 200 Teaching in Applied Mathematics (Fall 2026).

*Tutor and Reader* *Fall 2025 – Spring 2026*

- Tutor for AM 10 Mathematical Methods I (Linear Algebra), 3 consecutive quarters.
- Reader for CSE 16 Applied Discrete Mathematics; tutor for MATH 19B Calculus II.

## HONORS

Highest Honors in the Major, B.S. Applied Mathematics, UC Santa Cruz, 2026
Dean's Honors, UC Santa Cruz, 4 quarters

## SKILLS

**Languages:** Python (primary), C++ (working knowledge), MATLAB, R, Bash; familiar: RISC-V assembly
**ML and Scientific Computing:** PyTorch, JAX, NumPy, scikit-learn; familiar: Triton (GPU kernels)
**HPC and Tools:** Slurm, PBS, NCAR Derecho, multi-GPU H100 nodes, Linux, Git, SSH
**Mathematics:** Linear algebra, ODEs and PDEs, dynamical systems, numerical methods, probability and statistics

## COURSEWORK

**Graduate:** Scientific Machine Learning (AM 261); Applied Dynamical Systems (AM 224); Machine Learning and AI in Genomics (AM 234); Numerical Linear Algebra (AM 213A, in progress); Teaching in Applied Mathematics (AM 200, in progress)

**Mathematics and numerics:** Mathematical Methods I–III (AM 10, 20, 100); Intro to PDEs (AM 112); Dynamical Systems (AM 114); Scientific Computing (AM 129); Computational Methods (AM 147); Mathematical Modeling I–II (AM 170A/B); Applied Discrete Mathematics (CSE 16)

**Probability, statistics and finance:** Probability and Statistics (CSE 107); Classical and Bayesian Inference (STAT 132); Financial Mathematics (MATH 114)

**Computing:** GPU Programming (AM 148); Computer Systems and Assembly (CSE 12)

## ADDITIONAL INFORMATION

**Citizenship:** U.S. citizen

## REFERENCES

Available on request.